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Opening presenter and design lead · slides 1–5 · about 4:13 of narration

> THE PROJECT STORY
>
> We wanted to study dispatch decisions. April's data described aggregate patterns rather than individual journeys, so we built plausible evenings from evidence and explicit assumptions. We checked the implementation, replayed identical evenings under different transfer rules, and found a conditional trade-off between timely service and Lyft use. Our next work tests how robust those findings are.

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1 Your section

Slides 1–5: When should Night Ride move a waiting request to Lyft? → We narrowed the question to one controllable decision.

Purpose: Introduction, service context, proposal recap and refined research question.

You open the talk with slide 1's narration.

Your handoff sentence (end of slide 5): "That gives us the question. Miles will now explain how April's aggregate data became the ingredients of our scenarios."

2 Essential preparation reading

Read these in the shared guide (`01_Team_Presentation_Guide.pdf`, one folder up) and the methodology PDF:

- Slide 1 · When should Night Ride move a waiting request to Lyft?: guide, slide 1 (guide p. 14); methodology §1.1 The question, p. 6
- Slide 2 · The service and one request: guide, slide 2 (guide p. 15); methodology §1.3 System boundaries, p. 6; §4.6 Transfers and the Lyft service, p. 22
- Slide 3 · Where we started: guide, slide 3 (guide p. 16); methodology §1.4 From the September proposal to the current study, p. 8
- Slide 4 · Where we started: numbers we later corrected: guide, slide 4 (guide p. 17); methodology §1.4 From the September proposal to the current study, p. 8; §3.1 Volume: how many requests, p. 12
- Slide 5 · We narrowed the question to one controllable decision: guide, slide 5 (guide p. 18); methodology §1.4 From the September proposal to the current study, p. 8; §5.6 The dispatchers and the held-fixed choice, p. 31
- Everyone: guide sections “What every member must be able to explain” and “Quick Q&A navigation”.

3 Concepts you must explain confidently

- the service and one request’s journey (request, wait, curb arrival, boarding, drop-off; cancellation and no-show)
- why a Lyft order has a price (an extra trip Hopkins pays for; cost unknown)
- what the original proposal planned and what changed (trip records never arrived)
- the four corrected proposal numbers, especially the 30-day total
- why holding one dispatcher fixed is a scope choice, not a claim about synthetic data

4 Questions you lead on

Answer these first; route anything else to its owner (guide, Quick Q&A navigation).

- Why this project; what the service is — guide slides guide, slide 1 (guide p. 14), guide, slide 2 (guide p. 15)
- The original proposal and what changed — guide slides guide, slide 3 (guide p. 16), guide, slide 4 (guide p. 17); backups 38
- Why hold one dispatcher fixed (the pivot) — guide slides guide, slide 5 (guide p. 18); backups 38

5 Your narration (exact words)

Slide 1 · When should Night Ride move a waiting request to Lyft?

about 42 s · open slide

SAY (exact narration)

We began with a dispatch problem. The data led us first to a modelling problem. Good afternoon, we are Team FourSight. Our question is simple to state: when a rider has been waiting for a Night Ride van, when should that request be moved to Lyft instead? [optional] Today we will show what April's aggregate data let us build, how we turned it into constructed evenings, and what our first simulations revealed.

TRANSITION

First, the service itself, and one request.

Slide 2 · The service and one request

about 55 s · open slide

SAY (exact narration)

Homewood-area Night Ride pools requests in shared vans. A dispatcher can move waiting riders to Lyft. We are studying when that transfer should happen. Here is one request's journey: request, wait, the vehicle reaches the curb, boarding, drop-off. It breaks if the rider gives up while waiting, a cancellation, or is absent when the van arrives, a no-show after a three-minute wait. Four words matter later, and they are on the tiles: booking, Lyft order, curb arrival, and pickup, the rider boarding. Each Lyft order is an extra trip Hopkins pays for, so moving riders to Lyft has a price.

TRANSITION

Our September proposal approached this differently.

Slide 3 · Where we started

about 52 s · open slide

SAY (exact narration)

Our original plan was to reconstruct service nights and study dispatch decisions using an optimization model. These are two slides from that proposal. On the left, the data plan: aggregate dashboards were available, and de-identified trip-level extracts were coming, with request, pickup and drop-off times for each ride. Trip-level data would let us replay actual service nights. On the right, the method: compare fast heuristics, a rolling-horizon optimization model and a hybrid, all on the same request stream. Those trip-level extracts never arrived, so real nights could not be replayed.

TRANSITION

The proposal also contained numbers we later had to correct.

Slide 4 · Where we started: numbers we later corrected

about 50 s · open slide

SAY (exact narration)

Four numbers on these original slides were wrong or misread. One: “about 5,100 rides per hour at 6 pm” is a total over thirty days, so about 170 rides in one evening’s six o’clock hour. Two: “about 28 vans” is the dashboard’s all-services average, not a Homewood-only count. [optional] Three: the 7 pm bar shows 4,900; our digitized value is about 3,960. Four: “one in four trips overflow to Lyft” is really a share of riders: 24.5 percent of riders rode Lyft. These corrections shaped everything after.

TRANSITION

With aggregate data only, we narrowed the question.

Slide 5 · We narrowed the question to one controllable decision

about 54 s · open slide

SAY (exact narration)

We now hold one specified van dispatcher fixed and change only the Lyft transfer rule. This lets us study how the timing and information used by that rule change service and Lyft use. It is the question we chose; we don't know the internals of Hopkins' real dispatcher; and a controlled comparison needs everything else held fixed. Our van dispatcher re-plans every two minutes and at each new request, cancellation or no-show while waiting, or van coming on duty, modelled on the vendor TransLoc's published design from 2016. That is our modelling choice, not verified current Hopkins practice.

HANDOFF: Irfan Setiadi → Miles Zhou

That gives us the question. Miles will now explain how April's aggregate data became the ingredients of our scenarios.

6 Manual verification activity: Units and the proposal correction

> **WHAT HUMAN VERIFICATION MEANS**
>
> A program can generate and check an answer. Human verification means a teammate independently works through a defined example, records their calculation and compares it with the implementation.

This is a guided preparation activity, not a quiz or an exam. Work through the procedure on your own first and write down your result. Only then turn to the reference answer, which is on a separate page. If your result differs, record the difference rather than adjusting your working to match: a disagreement is useful information. The reference answer was produced with AI assistance from the project files; it does not count as a teammate's verification.

Related slides: Slides 3–4; also slide 7. **Goal:** Explain independently why "about 5,100 completed rides at 6 pm" summed over thirty days does not mean 5,100 rides in one evening's hour, and convert it to rides per evening.

Source inputs

Input	Value	Provenance
April dashboard chart "Completed rides by hour", 18:00 bar	30-day total	Chart subtitle: summed over the 30 days of April; filtered to Homewood Night Ride as printed
Tooltip value for that bar	5,098	Read on the dashboard (restricted source; number only)
Digitized value for that bar	5,089.09	Our pixel reading, project file `obs_p09_rides_by_hour.csv`
Proposal wording	"approximately 5,100 rides per hour"	September proposal, slide 7 (rounded, illustrative)
Days in April	30	Calendar
April status totals (context only)	50,297 requests; 24,095 completed	Dashboard status totals

Procedure

1. Write down, in words, what one bar of the chart counts: which rides, which hour, over how many days.
2. Divide the digitized value by the number of days. Repeat with the tooltip value and with the rounded 5,100. Keep two decimals for the exact values; round only at the end.
3. State the result as "completed rides in one evening's 6 pm hour", with units.

4. Explain in one or two sentences why “5,100 rides per hour” is a units error, not a small rounding difference.
5. List what this calculation establishes, and what it does *not* establish about (a) total requests, which include bookings that were not completed, and (b) a Homewood-specific load per vehicle.

Worksheet

YOUR WORKING

Step-by-step working (use the lines, or attach your own sheet)

My result

Record

Status	**Awaiting human completion**
Name	
Date completed	
My result	
Matches the reference? (yes / partly / no)	
Discrepancy, and what I did about it	
What remains unverified	

Reference answer

REFERENCE ANSWER (compare only after finishing your own working)

Calculation. 5,089.09 / 30 = 169.64; 5,098 / 30 = 169.93; 5,100 / 30 = 170.00. All three give about **170 completed rides in one April evening's 6 pm hour, on average**. The 5,089.09 and 5,098 are two readings of the same bar (digitized and tooltip); "about 5,100" and "about 170" are rounded, illustrative values.

Why "5,100 per hour" is wrong. The bar adds up the 6 pm hour over thirty evenings. Reading it as one hour overstates a single evening's hourly rides by a factor of 30, not by a rounding margin.

What it establishes. The average number of *completed* rides recorded in the 18:00–19:00 hour per April evening.

What it does not establish.

- *Total requests.* Completed rides are a subset of requests. Getting to about 355 requests needs the monthly ratio 50,297 / 24,095 = 2.087 requests per completed ride, and the further **assumption** that this monthly ratio holds within the hour (slide 7).
- *A Homewood-specific load per vehicle.* The rides chart is filtered to Homewood Night Ride as printed, but the vehicle count used next on slide 7 (27.6) is an "All services" figure, so the per-vehicle load is not Homewood-specific.
- *Night-to-night variation.* A 30-day average says nothing about how busy any one evening was.

Comparison checklist

- ☐ Did I treat the bar as a 30-day total, not one hour?
- ☐ Is my per-evening value within rounding of 170 for each of the three inputs?
- ☐ Did I keep the exact digitized and tooltip values separate from the rounded 5,100 and 170?
- ☐ Did I say that completed rides are not requests, and name the assumption needed to reach requests?
- ☐ Did I note that the per-vehicle load mixes a Homewood ride count with an "All services" vehicle count?

Record any disagreement in the table above. Do not change your own working to match the reference.

7 Rehearsal and readiness checklist

- ☐ Read my slides' preparation notes in the shared guide.
- ☐ Read my slides in the final deck (`02_Final_Presentation.pdf`) and know which backup slides I may need.
- ☐ Said my narration aloud with a stopwatch; my section took _____ (estimate 4:13).
- ☐ Practised my opening and handoff sentences with the neighbouring presenter.

☐ Can explain every concept listed above without the slides.

☐ Completed my verification activity and filled in the record (status changed from “Awaiting human completion” only by me).

☐ Took part in one timed full-team rehearsal (total time: _____).

8 Design lead: instructions for restyling

The editable deck is in the folder `designer_Irfan_Setiadi` (file `BlueJay_Midterm_v0.4.pptx`). The PDFs in this pack remain the reference for content.

- The presentation content is frozen. Restyling must not change meaning, numbers, rule names or slide order.
- Keep every factual sentence as written, including the lead sentences in blue and the statements of what a result does *not* show.
- Keep both label systems and their wording: evidence labels (OBSERVED, DERIVED, ASSUMED, DESIGNED, SIMULATED, ILLUSTRATIVE, and combinations such as DERIVED + ASSUMED) and execution-status bands (TESTED IN SIMULATION, PLANNED — NOT YET RUN, PROPOSED — NOT IMPLEMENTED). Do not rely on colour alone.
- Keep the component colours consistent: simulator grey, van optimizer blue, transfer rule orange, hidden information violet (dashed).
- Keep sources readable. Do not move essential qualifications (for example “curb arrival, not boarding”, “this measure favours early transfer”, “no noise margin applied”) into footnotes or small print.
- **The slide count and order are frozen** (28 main slides, the divider on slide 29, backups on 30–38): every slide number in the guide, the briefs and the Q&A table depends on them. Do not add, remove or reorder slides.
- Keep the original proposal slides framed and labelled ORIGINAL PROPOSAL, with the masked maps and the numbered correction markers on slide 4.
- Charts are native and editable; keep direct labels and a single value axis. Keep backup slides and their “back to slide” links working.
- Check slides 7, 14, 16 and 22–25 for text fit after any font change; the preview was rendered with LibreOffice.